Minjee Kim

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Research Profile

My research develops *sufficient dimension reduction* (SDR) methods, which reduce high-dimensional predictors to a few summaries that retain all information about the outcome. I extend SDR to settings where classical methods break down, including multivariate outcomes, censored survival times, and nonlinear relationships in functional data such as curves. Ultimately, I aim to establish dimension reduction as a reliable tool for interpreting complex biomedical data.

Education

Ph.D. in Statistics, Ewha Womans University — 2022 – 2027 (expected)
Advisor: Jae Keun Yoo (integrated M.S.–Ph.D. program)
Dissertation: *Sufficient Dimension Reduction for Nonstandard Responses*

Visiting Student, Mechanical & Industrial Engineering, University of Toronto — Jan – Jun 2025

B.S. in Architectural & Urban Systems Engineering, Ewha Womans University — 2017 – 2022
Double major in Statistics

Research Experience

Graduate Research Assistant, NRF-funded projects on sufficient dimension reduction — 2022 – 2027
PI: Jae Keun Yoo, Ewha Womans University

Visiting Researcher, Sunnybrook Health Sciences Centre (U of T CARTE Program) — Jan – Jun 2025
PI: Michael Hardisty, Sunnybrook Research Institute

Publications

Journal Articles — * equal contribution

[1] **Minjee Kim***, Minjeong Kim*, and Jae Keun Yoo (2026). Non-elliptical dimension reduction in survival regression. *Stat*, 15(3), e70162.

[2] **Minjee Kim***, Yujin Park*, Kyongwon Kim, and Jae Keun Yoo (2025). Nonlinear functional sufficient dimension reduction via principal fitted components. *Statistics and Computing*, 35(4), 95.

[3] **Minjee Kim** and Jae Keun Yoo (2025). Covariance projective resampling informative predictor subspace for multivariate regression. *Journal of the Korean Statistical Society*, 54(3), 920–931.

[4] **Minjee Kim** and Jae Keun Yoo (2024). Applications of response dimension reduction in large p-small n problems. *Communications for Statistical Applications and Methods*, 31(2), 191–202.

Conference Proceedings

[5] Eunseon Seong, **Minjee Kim**, Dareen Eom, Gyeonghwa Heo, Urban Emmenegger, Saleh Tabatabaei, Tayler Declan Ross, and Michael Hardisty (2026). Dynamic mortality prediction for prostate cancer based on irregularly sampled multimodal EHR and musculoskeletal image-based biomarkers. *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*, 399–410.

Manuscripts

[1] **Minjee Kim** and Jae Keun Yoo. Second-order covariance projective resampling for sufficient dimension reduction with multivariate responses. *Under review at Statistics and Computing.*

[2] **Minjee Kim** and Jae Keun Yoo. Projective-resampling sliced mean regression. *In preparation.*

Presentations

Invited: Junior Statisticians' Paper Presentation, WISK, Korean Statistical Society — Oct 2026
Nonlinear functional SDR via principal fitted components

13th Conference of the IASC Asian Regional Section (IASC-ARS) — Dec 2025
Covariance projective resampling informative predictor subspace for multivariate regression

8th International Conference on Econometrics and Statistics (EcoSta) — Aug 2025
Nonlinear functional SDR via principal fitted components

Interim Conference of the IASC Asian Regional Section (IASC-ARS) — Dec 2024
Applications of response dimension reduction in large p-small n problems

Winter Conference of the Korean Data Analysis Society — Dec 2024
Applications of response dimension reduction in large p-small n problems

Teaching Experience

Teaching Assistant, Department of Statistics, Ewha Womans University — 2022 – 2026

- *Topics in Multivariate Analysis* (graduate): guest lecture
- *Regression Analysis*, *Bayesian Statistics*: R lab sessions, tutoring
- *Statistics II*, *Mathematical Statistics*: grading and solutions

Workshop Assistant, Ewha GSIS Computational Social Science Workshop — 2022, 2023

- English-language R support in hands-on sessions

Honors & Awards

Best Paper Award, Ewha Womans University — 2025
Ewha Excellent Science Scholar Fellowship, Ewha Womans University — 2022 – 2023

Professional Service

Session Volunteer, 33rd International Biometric Conference (IBC) — Jul 2026
Session Chair, 13th Conference of the IASC Asian Regional Section (IASC-ARS) — Dec 2025

Skills

Computing: R (primary), Python and SQL, LaTeX
Languages: English (professional), Korean (native)